

Table 1: Security model terms and their definitions

Term	Definition
H2O	A collection of H2O nodes that work together. In the H2O Flow Web UI, the cluster status menu item shows the list of nodes in a H2O cluster.
H2O node	One VM instance running the H2O main class. One H2O node corresponds to one OS-level process. In the YARN case, one H2O node corresponds to one mapper instance and one YARN container.
H2O embedded Web port	Each H2O node contains an embedded Web port (by default port 54321). This Web port hosts H2O Flow as well as the H2O REST API. The user interacts directly with this Web port.
H2O internal communication port	Each H2O node also has an internal port (Web port+1, so by default port 54322) for internal node-to-node communication. This is a proprietary binary protocol. An attacker using a tool like tcpdump or Wireshark may be able to reverse engineer data captured on this communication path.

File security in H2O: H2O is a typical user prototype. To ensure user information safety for H2O, there's nothing to be done specifically. HDFS permissions and the operating system just work well to ensure security.

H2O Flow

H2O is an interactive Web based framework which allows users to merge text, plots, code execution, mathematics and rich media in a single document.

By using H2O Flow, one can present, annotate, rerun and share one's workflow. It also allows users to build prototypes, to import files, and frequently improve them. Based on user prototypes, one can make projections and join rich text to generate vignettes of your work — all within Flow's browser based domain.

With a modern graphical interaction, Flow's hybrid user interface seamlessly merges command line computing. Rather than showing the outcome as plain text, Flow delivers a point and click user interaction for each H2O performance. It allows users to approach any H2O object by means of tabular information.

H2O is an open source predictive analytics program and also provides heterogeneous conventional analytics mechanisms. It enables the collaboration of exceptional mathematics and giant, production-grade parallel processing with unequalled ease of use. Data scientists and business analysts prefer H2O for scalable and fast machine learning. **END** 

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